

# CONSTRUCTION ENGINEERING

The Construction Engineering Program is accredited by the Engineering Accreditation Commission of ABET, <https://www.abet.org> (<https://www.abet.org/>), under the commission's General Criteria and Program Criteria for Construction and Similarly Named Engineering Programs.

The process of construction involves the organization, administration, and coordination of labor resource requirements, temporary and permanent materials, equipment, supplies and utilities, money, technology and methods. These must be integrated in the most efficient manner possible to complete construction projects on schedule, within the budget, and according to the standards of quality and performance specified by the project owner or designer. The Construction Engineering curriculum blends engineering, management, and construction sciences into a study of the processes of construction whereby designer's plans and specifications are converted into physical structures and facilities. The curriculum develops the ability of students to be team workers, creative thinkers, and effective communicators. This is achieved by encouraging students to:

- Interact with practicing professionals,
- Gain work experience during summer jobs, internship, and cooperative education assignments that emphasize the knowledge required of construction engineers.
- Develop leadership skills by participating in student organizations.
- Develop, analyze, and interpret alternative solutions to open-ended problems.

Construction engineers need to possess teamwork skills and strong fundamental knowledge of engineering design and management principles, including knowledge of business procedures, economics, and human behavior. Graduates of this curriculum may expect to engage in design of temporary structures, coordination of project design, systems design, cost estimating, planning and scheduling, company and project management, materials procurement, equipment selection, and cost control. With the emergence of integrated project delivery methods such as design-build construction, the role of the construction engineer is expanding the need for trained professionals that understand both aspects of the project delivery environment. The curriculum offers opportunities to study options concerned with building, electrical, infrastructure, or mechanical construction.

## Student Learning Outcomes

At graduation students of the construction engineering curriculum will have:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences.
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

## Program Educational Objectives:

By three to five years after graduation, graduates of the civil, construction, and environmental engineering programs will have

1. Pursued successful careers and expertise in civil engineering, construction engineering, environmental engineering, or a related profession that includes professionalism and increasing responsibility.
2. Collaborated effectively on multi-disciplinary teams to address the needs of society and the environment
3. Pursued lifelong learning, professional development, and licensure as appropriate for their career goals and for developments in the field.

Qualified construction engineering students within 30 credits of completing their degree may apply for concurrent enrollment in the Graduate College. See Civil Engineering (<http://www.ccee.iastate.edu/academics/graduate/>) Graduate Study for more information.

Students who successfully complete the Construction Engineering curriculum will be prepared for entry into the field or for further study at the graduate level in construction engineering or related fields of study, such as law, business, and/or other engineering disciplines.

The curriculum in Construction Engineering, leading to a Bachelor of Science degree can be referenced here: [www.catalog.iastate.edu/collegeofengineering/constructionengineering/#curriculumtext](http://www.catalog.iastate.edu/collegeofengineering/constructionengineering/#curriculumtext).

## Curriculum in Construction Engineering

Administered by the Department of Civil, Construction and Environmental Engineering; leading to the degree bachelor of science.

Construction Engineering students will initially be considered construction undecided option when starting the degree and are encouraged to choose an option in Construction Engineering before their 5th semester. If not, option courses taken by the student will not be applied until the student chooses an option. Students must pick an option prior to graduation. The four options in the program are building, electrical, infrastructure, and mechanical. The Core courses for each option are the same whereas the option selections of the degree audit will be comprised of different classes that explore the particular focus the students have chosen.

### Total credits required: Building Option -128.0, Electrical Option - 127.0, Infrastructure Option - 127.0, Mechanical Option - 127.0 cr.

The Construction Engineering program requires a grade of a C or better for any transfer credit course that is applied to the degree program (but will not be calculated into the ISU cumulative GPA, Basic Program GPA or Core GPA). Note: Department does not allow Pass/Not Pass credits to be used to meet graduation requirements for either required or elective courses.

#### International Perspectives: 3 cr. <sup>1</sup>

#### U.S. Cultures and Communities: 3 cr. <sup>1</sup>

#### Communication Proficiency/Library requirements:

|                                                                                         |                                                                                            |   |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---|
| ENGL 1500                                                                               | Critical Thinking and Communication (Must have a C or better in this course)               | 3 |
| ENGL 2500                                                                               | Written, Oral, Visual, and Electronic Composition (Must have a C or better in this course) | 3 |
| LIB 1600                                                                                | Introduction to College Level Research                                                     | 1 |
| Business Communication Elective: one course of the following with a minimum grade of C. |                                                                                            | 3 |
| ENGL 3020                                                                               | Business Communication                                                                     |   |
| ENGL 3090                                                                               | Proposal and Report Writing                                                                |   |
| ENGL 3140                                                                               | Technical Communication                                                                    |   |

**Total Credits** 10

#### Social Sciences and Humanities: 12 cr.

|                                             |           |
|---------------------------------------------|-----------|
| Social Sciences and Humanities <sup>2</sup> | 6         |
| International Perspectives <sup>1</sup>     | 3         |
| U.S. Cultures and Communities <sup>1</sup>  | 3         |
| <b>Total Credits</b>                        | <b>12</b> |

#### Basic Program: 24 cr. <sup>3</sup>

Minimum GPA of 2.00 required for this set of courses to graduate, (please note that transfer course grades will not be calculated into the Basic Program GPA).<sup>3</sup>

|                      |                                                                                                                        |           |
|----------------------|------------------------------------------------------------------------------------------------------------------------|-----------|
| CHEM 1670            | General Chemistry for Engineering Students                                                                             | 4         |
| or CHEM 1770         | General Chemistry I                                                                                                    |           |
| ENGL 1500            | Critical Thinking and Communication (Must have a C or better in this course)                                           | 3         |
| ENGR 1010            | Engineering Orientation                                                                                                | R         |
| CE 1600              | Systematic Problem Solving and Computer Applications for Civil, Construction, and Environmental Engineers <sup>3</sup> | 3         |
| LIB 1600             | Introduction to College Level Research                                                                                 | 1         |
| MATH 1650            | Calculus I                                                                                                             | 4         |
| MATH 1660            | Calculus II                                                                                                            | 4         |
| PHYS 2310            | Introduction to Classical Physics I                                                                                    | 4         |
| PHYS 2310L           | Introduction to Classical Physics I Laboratory                                                                         | 1         |
| <b>Total Credits</b> |                                                                                                                        | <b>24</b> |

#### Math and Physical Science: 12 cr.

|                      |                                                          |              |
|----------------------|----------------------------------------------------------|--------------|
| STAT 3005            | Engineering Statistics                                   | 3-4          |
| or STAT 3031         | Probability and Statistical Inference for Engineers      |              |
| MATH 2650            | Calculus III                                             | 3-4          |
| or MATH 2070         | Matrices and Linear Algebra                              |              |
| MATH 2670            | Elementary Differential Equations and Laplace Transforms | 4            |
| PHYS 2320            | Introduction to Classical Physics II                     | 4            |
| PHYS 2320L           | Introduction to Classical Physics II Laboratory          | 1            |
| <b>Total Credits</b> |                                                          | <b>15-17</b> |

#### Construction Engineering Core: 27 cr. Minimum 2.00 GPA for this set of courses to graduate (please note that transfer course grades will not be calculated into the Core GPA):

|            |                                                   |   |
|------------|---------------------------------------------------|---|
| CE 2710    | Engineering Foundations of Statics                | 3 |
| & CE 2720  | and Applied Engineering Statics                   |   |
| or CE 2740 | Engineering Statics                               |   |
| CONE 2410  | Construction Materials and Methods                | 3 |
| CE 3320    | Structural Analysis I                             | 3 |
| EM 3240    | Mechanics of Materials                            | 3 |
| CONE 4220  | Construction Cost Estimating and Cost Engineering | 3 |
| CONE 4410  | Construction Planning, Scheduling, and Control    | 3 |
| ABE 3780   | Mechanics of Fluids                               | 3 |
| CONE 4870  | Construction Engineering Design I                 | 3 |

|                      |                                    |           |
|----------------------|------------------------------------|-----------|
| CONE 4880            | Construction Engineering Design II | 3         |
| <b>Total Credits</b> |                                    | <b>27</b> |

**Select remaining courses from one of the following options:**  
**Building Option 27 cr:**

|                                          |                                                       |           |
|------------------------------------------|-------------------------------------------------------|-----------|
| CE 3600                                  | Geotechnical Engineering                              | 4         |
| CONE 3220                                | Construction Equipment and Heavy Construction Methods | 3         |
| CONE 3400                                | Concrete and Steel Construction                       | 3         |
| CE 3330                                  | Structural Steel Design I                             | 3         |
| CE 3340                                  | Reinforced Concrete Design I                          | 3         |
| CE 3830                                  | Design of Portland Cement Concrete                    | 1         |
| CONE 3520                                | Mechanical Systems in Buildings                       | 3         |
| CONE 3530                                | Electrical Systems in Buildings                       | 3         |
| EM 3270                                  | Mechanics of Materials Laboratory                     | 1         |
| Engineering Topics Elective <sup>2</sup> |                                                       | 3         |
| <b>Total Credits</b>                     |                                                       | <b>27</b> |

**Electrical Option 26 cr:**

|                                          |                                      |           |
|------------------------------------------|--------------------------------------|-----------|
| EE 2010                                  | Electric Circuits                    | 4         |
| EE 2300                                  | Electronic Circuits and Systems      | 4         |
| EE 3030                                  | Energy Systems and Power Electronics | 3         |
| CONE 3520                                | Mechanical Systems in Buildings      | 3         |
| CONE 3530                                | Electrical Systems in Buildings      | 3         |
| EE 4560                                  | Power System Analysis I              | 3         |
| EE 4570                                  | Power System Analysis II             | 3         |
| Engineering Topics Elective <sup>2</sup> |                                      | 3         |
| <b>Total Credits</b>                     |                                      | <b>26</b> |

**Infrastructure Option 26 cr:**

|                                          |                                                       |           |
|------------------------------------------|-------------------------------------------------------|-----------|
| CE 3600                                  | Geotechnical Engineering                              | 4         |
| CONE 3220                                | Construction Equipment and Heavy Construction Methods | 3         |
| CONE 3400                                | Concrete and Steel Construction                       | 3         |
| CE 3330                                  | Structural Steel Design I                             | 3         |
| CE 3340                                  | Reinforced Concrete Design I                          | 3         |
| CE 3820                                  | Design of Concretes                                   | 3         |
| EM 3270                                  | Mechanics of Materials Laboratory                     | 1         |
| Engineering Topics Elective <sup>2</sup> |                                                       | 6         |
| <b>Total Credits</b>                     |                                                       | <b>26</b> |

**Mechanical Option 26 cr:**

|           |                                 |   |
|-----------|---------------------------------|---|
| ME 2310   | Engineering Thermodynamics I    | 3 |
| ME 4360   | Heat Transfer                   | 4 |
| CONE 3520 | Mechanical Systems in Buildings | 3 |
| CONE 3530 | Electrical Systems in Buildings | 3 |

|                                          |                                                            |           |
|------------------------------------------|------------------------------------------------------------|-----------|
| EE 4420                                  | Introduction to Circuits and Instruments                   | 2         |
| EE 4480                                  | Introduction to AC Circuits and Motors                     | 2         |
| ME 4410                                  | Fundamentals of Heating, Ventilating, and Air Conditioning | 3         |
| ME 4420                                  | Heating and Air Conditioning Design                        | 3         |
| Engineering Topics Elective <sup>2</sup> |                                                            | 3         |
| <b>Total Credits</b>                     |                                                            | <b>26</b> |

**Additional Required Courses: 26 cr.**

|                                                      |                                                                                            |           |
|------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------|
| CONE 1210                                            | Cornerstone Learning Community: Orientation to Academic Life                               | 1         |
| CONE 1220                                            | Cornerstone Learning Community: Orientation to Professional Life                           | 1         |
| CONE 2220                                            | Contractor Organization and Management of Construction                                     | 3         |
| CONE 2510                                            | Mechanical/Electrical Materials and Methods                                                | 1         |
| CE 1110                                              | Geomatics                                                                                  | 3         |
| CE 1700                                              | Graphics for Civil Engineering                                                             | 2         |
| IE 3050                                              | Engineering Economic Analysis                                                              | 3         |
| ENGL 2500                                            | Written, Oral, Visual, and Electronic Composition (Must have a C or better in this course) | 3         |
| Law Elective                                         |                                                                                            | 3         |
| CONE 3800                                            | Engineering Law                                                                            |           |
| ACCT 2150                                            | Legal Environment of Business                                                              |           |
| Business Communication Elective (minimum grade of C) |                                                                                            | 3         |
| ENGL 3020                                            | Business Communication                                                                     |           |
| ENGL 3090                                            | Proposal and Report Writing                                                                |           |
| ENGL 3140                                            | Technical Communication                                                                    |           |
| <b>Total Credits</b>                                 |                                                                                            | <b>23</b> |

**Co-op/Internships - Optional**

1. These university requirements will add to the minimum credits of the program unless the university-approved courses are also approved by the department to meet other course requirements within the degree program. U.S. Cultures and Communities and International Perspectives courses may not be taken Pass/Not Pass.
2. Choose from program approved list (<https://www.ccee.iastate.edu/undergraduate-programs/construction-engineering/construction-engineering-forms-and-information-by-cohort/>). Note: 3 of the 6 Social Sciences and Humanities credits must be ECON 1010 or ECON 1020.
3. See Basic Program for Engineering Curricula (<https://catalog.iastate.edu/collegeofengineering/#basicprogramcurriculertext>) for accepted substitutions for curriculum designated courses in the Basic Program <https://>

[www.engineering.iastate.edu/classification/students/basic-program/](http://www.engineering.iastate.edu/classification/students/basic-program/).

See also: A 4-year plan of study grid showing course template by semester for a building option in Construction Engineering.

See also: A 4-year plan of study grid showing course template by semester for an electrical option in Construction Engineering.

See also: A 4-year plan of study grid showing course template by semester for a infrastructure option in Construction Engineering.

See also: A 4-year plan of study grid showing course template by semester for a mechanical option in Construction Engineering.

Construction Engineering, B.S. Building option

#### First Year

| Fall      | Credits Spring      | Credits   |
|-----------|---------------------|-----------|
| CONE 1210 | 1 CONE 1220         | 1         |
| ENGR 1010 | R CE 1700           | 2         |
| MATH 1650 | 4 MATH 1660         | 4         |
| CHEM 1670 | 4 PHYS 2310         | 4         |
| CE 1600   | 3 PHYS 2310L        | 1         |
| ENGL 1500 | 3 ENGL 2500         | 3         |
| LIB 1600  | 1 ECON 1010 or 1020 | 3         |
| <b>16</b> |                     | <b>18</b> |

#### Second Year

| Fall              | Credits Spring | Credits   |
|-------------------|----------------|-----------|
| CONE 2220         | 3 CONE 2410    | 3         |
| CE 1110           | 3 CONE 2510    | 1         |
| MATH 2650 or 2070 | 3-4 MATH 2670  | 4         |
| PHYS 2320         | 4 CE 2720      | 2         |
| PHYS 2320L        | 1 IE 3050      | 3         |
| CE 2710           | 1 SSH Elective | 3         |
| <b>15-16</b>      |                | <b>16</b> |

#### Third Year

| Fall              | Credits Spring           | Credits   |
|-------------------|--------------------------|-----------|
| CONE 3220         | 3 CONE 3400              | 3         |
| CONE 3520         | 3 CONE 3800 or ACCT 2150 | 3         |
| CONE 3530         | 3 CE 3600                | 4         |
| EM 3240           | 3 CE 3320                | 3         |
| STAT 3005 or 3031 | 3-4 EM 3270              | 1         |
|                   | ABE 3780                 | 3         |
| <b>15-16</b>      |                          | <b>17</b> |

#### Fourth Year

| Fall                        | Credits Spring              | Credits   |
|-----------------------------|-----------------------------|-----------|
| CONE 4220                   | 3 CONE 4870                 | 3         |
| CONE 4410                   | 3 CONE 4880                 | 3         |
| CE 3830                     | 1 ENGL 3020, 3090, or 3140  | 3         |
| CE 3330                     | 3 CE 3340                   | 3         |
| Engr Topics Elective        | 3 International Perspective | 3         |
| US Cultures and Communities | 3                           |           |
| <b>16</b>                   |                             | <b>15</b> |

Construction Engineering, B.S. Electrical option

#### First Year

| Fall      | Credits Spring      | Credits   |
|-----------|---------------------|-----------|
| CONE 1210 | 1 CONE 1220         | 1         |
| ENGR 1010 | R CE 1700           | 2         |
| MATH 1650 | 4 MATH 1660         | 4         |
| CHEM 1670 | 4 PHYS 2310         | 4         |
| CE 1600   | 3 PHYS 2310L        | 1         |
| ENGL 1500 | 3 ENGL 2500         | 3         |
| LIB 1600  | 1 ECON 1010 or 1020 | 3         |
| <b>16</b> |                     | <b>18</b> |

#### Second Year

| Fall              | Credits Spring | Credits   |
|-------------------|----------------|-----------|
| CONE 2220         | 3 CONE 2410    | 3         |
| CE 1110           | 3 CONE 2510    | 1         |
| MATH 2650 or 2070 | 3-4 MATH 2670  | 4         |
| PHYS 2320         | 4 CE 2720      | 2         |
| PHYS 2320L        | 1 IE 3050      | 3         |
| CE 2710           | 1 SSH Elective | 3         |
| <b>15-16</b>      |                | <b>16</b> |

#### Third Year

| Fall              | Credits Spring                  | Credits   |
|-------------------|---------------------------------|-----------|
| CONE 3520         | 3 CONE 3800 or ACCT 2150        | 3         |
| CONE 3530         | 3 EE 2300                       | 4         |
| EE 2010           | 4 EE 3030                       | 3         |
| EM 3240           | 3 ABE 3780                      | 3         |
| STAT 3005 or 3031 | 3-4 US Cultures and Communities | 3         |
| <b>16-17</b>      |                                 | <b>16</b> |

#### Fourth Year

| Fall      | Credits Spring | Credits |
|-----------|----------------|---------|
| CONE 4220 | 3 CONE 4870    | 3       |

|                             |   |                           |   |
|-----------------------------|---|---------------------------|---|
| CONE 4410                   | 3 | CONE 4880                 | 3 |
| EE 4560                     | 3 | EE 4570                   | 3 |
| CE 3320                     | 3 | ENGL 3020, 3090, or 3140  | 3 |
| Engineering Topics Elective | 3 | International Perspective | 3 |
| <b>15</b>                   |   | <b>15</b>                 |   |

## Construction Engineering, B.S. Infrastructure option

**First Year**

| <b>Fall</b> | <b>Credits</b> | <b>Spring</b>     | <b>Credits</b> |
|-------------|----------------|-------------------|----------------|
| CONE 1210   | 1              | CONE 1220         | 1              |
| ENGR 1010   | 2              | R CE 1700         | 2              |
| MATH 1650   | 4              | MATH 1660         | 4              |
| CHEM 1670   | 4              | PHYS 2310         | 4              |
| CE 1600     | 1              | PHYS 2310L        | 1              |
| ENGL 1500   | 3              | ENGL 2500         | 3              |
| LIB 1600    | 3              | ECON 1010 or 1020 | 3              |
| <b>16</b>   |                | <b>18</b>         |                |

**Second Year**

| <b>Fall</b>       | <b>Credits</b> | <b>Spring</b> | <b>Credits</b> |
|-------------------|----------------|---------------|----------------|
| CONE 2220         | 3              | CONE 2410     | 3              |
| CE 1110           | 1              | CONE 2510     | 1              |
| MATH 2650 or 2070 | 4              | MATH 2670     | 4              |
| PHYS 2320         | 2              | CE 2720       | 2              |
| PHYS 2320L        | 3              | IE 3050       | 3              |
| CE 2710           | 3              | SSH Elective  | 3              |
| <b>15-16</b>      |                | <b>16</b>     |                |

**Third Year**

| <b>Fall</b>            | <b>Credits</b> | <b>Spring</b>               | <b>Credits</b> |
|------------------------|----------------|-----------------------------|----------------|
| CONE 3220              | 3              | CONE 3400                   | 3              |
| CONE 3800 or ACCT 2150 | 3              | CE 3820                     | 3              |
| ABE 3780               | 4              | CE 3600                     | 4              |
| EM 3240                | 3              | CE 3320                     | 3              |
| STAT 3005 or 3031      | 1              | EM 3270                     | 1              |
|                        | 3              | US Cultures and Communities | 3              |
| <b>15-16</b>           |                | <b>17</b>                   |                |

**Fourth Year**

| <b>Fall</b>                 | <b>Credits</b> | <b>Spring</b>               | <b>Credits</b> |
|-----------------------------|----------------|-----------------------------|----------------|
| CONE 4220                   | 3              | CONE 4870                   | 3              |
| CONE 4410                   | 3              | CONE 4880                   | 3              |
| Engineering Topics Elective | 3              | Engineering Topics Elective | 3              |
| CE 3330                     | 3              | CE 3340                     | 3              |

|                           |   |                          |   |
|---------------------------|---|--------------------------|---|
| International Perspective | 3 | ENGL 3020, 3090, or 3140 | 3 |
| <b>15</b>                 |   | <b>15</b>                |   |

## Construction Engineering, B.S. Mechanical option

**First Year**

| <b>Fall</b> | <b>Credits</b> | <b>Spring</b>     | <b>Credits</b> |
|-------------|----------------|-------------------|----------------|
| CONE 1210   | 1              | CONE 1220         | 1              |
| ENGR 1010   | 2              | R CE 1700         | 2              |
| MATH 1650   | 4              | MATH 1660         | 4              |
| CHEM 1670   | 4              | PHYS 2310         | 4              |
| CE 1600     | 1              | PHYS 2310L        | 1              |
| ENGL 1500   | 3              | ENGL 2500         | 3              |
| LIB 1600    | 3              | ECON 1010 or 1020 | 3              |
| <b>16</b>   |                | <b>18</b>         |                |

**Second Year**

| <b>Fall</b>       | <b>Credits</b> | <b>Spring</b> | <b>Credits</b> |
|-------------------|----------------|---------------|----------------|
| CONE 2220         | 3              | CONE 2410     | 3              |
| CE 1110           | 1              | CONE 2510     | 1              |
| MATH 2650 or 2070 | 4              | MATH 2670     | 4              |
| PHYS 2320         | 2              | CE 2720       | 2              |
| PHYS 2320L        | 3              | IE 3050       | 3              |
| CE 2710           | 3              | SSH Elective  | 3              |
| <b>15-16</b>      |                | <b>16</b>     |                |

**Third Year**

| <b>Fall</b>       | <b>Credits</b> | <b>Spring</b>               | <b>Credits</b> |
|-------------------|----------------|-----------------------------|----------------|
| CONE 3520         | 3              | CONE 3800 or ACCT 2150      | 3              |
| CONE 3530         | 2              | EE 4420                     | 2              |
| ME 2310           | 2              | EE 4480                     | 2              |
| EM 3240           | 3              | CE 3320                     | 3              |
| STAT 3005 or 3031 | 3              | ABE 3780                    | 3              |
|                   | 3              | US Cultures and Communities | 3              |
| <b>15-16</b>      |                | <b>16</b>                   |                |

**Fourth Year**

| <b>Fall</b>                 | <b>Credits</b> | <b>Spring</b>             | <b>Credits</b> |
|-----------------------------|----------------|---------------------------|----------------|
| CONE 4220                   | 3              | CONE 4870                 | 3              |
| CONE 4410                   | 3              | CONE 4880                 | 3              |
| ME 4360                     | 3              | ME 4420                   | 3              |
| ME 4410                     | 3              | ENGL 3020, 3090, or 3140  | 3              |
| Engineering Topics Elective | 3              | International Perspective | 3              |
| <b>16</b>                   |                | <b>15</b>                 |                |

A concurrent bachelor of science/master of science (B.S./M.S.) degree program is available to qualified seniors at Iowa State University.

Civil Engineering, Graduate Programs, <https://www.ccee.iastate.edu/academics/graduate/>.

Go to <https://www.ccee.iastate.edu/prospective-graduate-students/concurrent-programs/> for further information.

## Graduate Study

An area of specialization in construction engineering and management is offered within the graduate program of the Department of Civil, Construction and Environmental Engineering. This specialization focuses on project management including and beyond the traditional iron triangle of scope, technical, and schedule to include context and financing, enabling project management of more complex projects. Three graduate degrees including, Master of Engineering (30 credits), Master of Science (30 credits), and Doctor of Philosophy (72 credits) are offered. The Master of Engineering degree is a coursework only option and the other degree programs require a research component at a level adjusted to the degree sought. All degrees are offered on-campus. A Master of Engineering may be completed off-campus through distance education. Other graduate degrees may potentially also be completed off-campus. All degrees require CE 5010, CE 5020, CE 5030. Students should refer to the CCEE Graduate Student Handbook for additional requirements for each graduate degree. Course options include but are not limited to:

|          |                                                                                                            |   |
|----------|------------------------------------------------------------------------------------------------------------|---|
| CE 5010  | Preconstruction Project Engineering and Management                                                         | 3 |
| CE 5020  | Construction Project Engineering and Management                                                            | 3 |
| CE 5030  | Construction Finance and Business Management                                                               | 3 |
| CE 5940A | Special Topics Construction Engineering and Mgt.: Planning and Scheduling                                  | 3 |
| CE 5940S | Special Topics Construction Engineering and Mgt.: Building Information Modeling                            | 3 |
| CE 5940U | Spl Topics Construction Engineering and Mgt.: Introduction to Utility Engineering and Coordination         | 3 |
| CE 5940Y | Special Topics in Construction Engineering and Management: Engineering for Disaster and Climate Resilience | 3 |

Undergraduate students may also qualify for the concurrent Bachelor of Science/Master of Science (B.S./M.S.) degree program. Courses are offered for minor work to students taking major work in other curricula or in interdepartmental programs.

A graduate certificate is also available which requires 12 credits of coursework. Courses required for the certificate are CE 5010, CE 5020, and CE 5030. The certificate may be completed on-campus or off-campus through distance education. For additional information see